

What to do when an error message appears

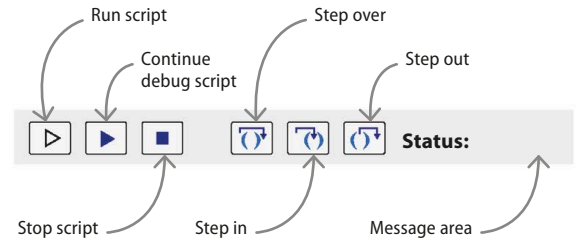
When an error message appears, the first task for programmers is to locate the bug. The compiler usually indicates which line of code caused a crash. However, some errors have a trickle-down effect and the actual error is several lines higher.

Error message checklist

- Review code for syntax errors. Go through the logic of the section to spot mistakes.
- If the error message doesn't make sense to you, try to find a solution online.
- Add print statements to code to display variables. A print statement is any command used to display text (including variables) to the user. Usually they appear in the programmer's console.
- Execute code and check that each value displayed in the console is correct.

Debugging

A debugger is a program used to find bugs in other programs. Most debuggers can run through the script (the program's instructions) in a step-by-step mode to isolate the source of the problem. Some debuggers can then fix the problem, or offer ways in which this can be done. The program can then be run again to see if the debugger has fixed the problems it found.



△ Breakpoints

A program freezes when it reaches a breakpoint, allowing programmers to detect the errors at their leisure. They can also check through the code one line at a time.

▽ Runtime errors

A runtime error is a specific type of logic error that occurs in the middle of a working program and causes it to crash. Usually, the program freezes or a pop-up box appears.

"=" instead of "=="
Mixing up "=" (assigns a value) and "==" (used for comparison) can lead to unpredictable results.

Incorrect Boolean expressions
Using AND, OR, and NOT incorrectly can cause the program to loop forever, or execute the wrong code.

Division by 0
When faced with this operation, programs usually crash.

Using different data types in the same operation
Mixing up text and numbers confuses a program, leading to an instant crash.

Segfault
A segmentation fault occurs when trying to access a restricted segment of memory.